Journal of Analytical and Applied Pyrolysis 186 (2025) 106919

Contents lists available at ScienceDirect

Journal of Analytical and Applied Pyrolysis

journal homepage: www.elsevier.com/locate/jaap

Influence of extractives on the composition of bio-oil from biomass pyrolysis – A review

Thamyris Q.S. Sampaio [a], Sirlene B. Lima [a,b], Carlos A.M. Pires [a,*]

[a] Polytechnic School, Federal University of Bahia, 2 R. Prof. Arístides Novis, Salvador, BA 40210-630, Brazil

[b] Interdisciplinary Bachelor's Degree in Environmental Sciences, Federal University of Bahia Recôncavo da Bahia, R. Rui Barbosa, Cruz das Almas, BA 44380-000

ARTICLE INFO

Keywords:
Pyrolysis
Extractives
Bio-oil
Desvolatilization
Lignocellulosic biomass
Constituent

ABSTRACT

Fast pyrolysis of biomass is a promising method for producing bio-oil, meeting the growing demand for renewable fuels with low environmental impact. However, the complexity of the reactions involved and the limited knowledge about the influence of extractives (proteins, starch, waxes, fats, resins, tannic acid, pigments, sugars, pectin, saponins, essential oils, etc.) on the pyrolysis of the main biomass components (cellulose, hemicellulose and lignin) make it difficult to formulate effective mathematical models to optimize the reaction system and scale the process commercially. This article provides a comprehensive overview of the state of the art regarding the influence of extractives on the pyrolysis of cellulose, hemicellulose, and lignin, as well as on the composition of bio-oil. The text critically discusses the characteristics of biomass and the fast pyrolysis of each of its macromolecules and extractives, establishing a reference base for studying the effects of extractives in this process. Additionally, it offers an updated perspective on how extractives affect the devolatilization of macromolecules, considering factors such as reaction activation, liquid yield, and types of char produced. The influence of extractives on the pyrolysis of each macromolecule and the distribution of chemical species present in bio-oil is also addressed, contributing to a better understanding of the process.

1. Introduction

Bio-oil, biochar, and non-condensable gases obtained from biomass pyrolysis in the absence of oxygen are energy alternatives and sources of chemical species that can help reduce pollutant emissions on our planet, as they originate from renewable sources [1–3]. Bio-oil's physical/chemical composition depends on biomass characteristics, such as macromolecular composition (cellulose, hemicellulose, lignin), particle size and shape, moisture content, and ash content. Factors such as reactor type, operating conditions (temperature, residence time, heating rate, and pressure), and cooling rate also play a role [4,5]. Bio-oil has a water content that ranges from 15 % to 30 % [4,6], and its oily fraction contains hundreds of compounds (300−400) [7,8], including acids, alcohols, ketones, aldehydes, phenols, ethers, esters, sugars, furans, nitrogen compounds, and multifunctional compounds [9,10]. Therefore, bio-oils have a high content of oxygenated compounds (35–40 %), which gives them a pH< 3 [6,11] and a lower higher heating value (HHV) compared to diesel and gasoline (17–20 MJ/kg) [6,12].

The most commonly used reactor configuration for this transformation is the fluidized bed reactor [1,13,14], but special reactors, such as vortex reactors, rotating blades reactors, rotating cone reactors, cyclone reactors, transported bed reactors, vacuum reactors, and auger reactors, can also be used [15,16]. These reactors can favor producing solids when biomass is heated slowly to a low temperature (330–400 °C) [17,18]. However, when the goal is to produce bio-oil, the reactors are operated at moderate temperatures (400–550 °C), with heating rates above 1000 °C/s, at atmospheric pressure [19], and with the residence time of molecules in the reactor of less than 2 seconds [1], whose process is called fast pyrolysis.

Fast pyrolysis is an advanced process with controlled parameters that offers high bio-oil yields. This, in turn, is a promising liquid fuel source for heat, power, and transportation generation, in addition to high-value chemical species [20–22]. Em geral, a biomassa passa por quatro estágios distintos na ausência de oxigênio: 1) aquecimento inicial, onde a umidade livre e a água fracamente ligada ao material são liberadas; 2) desidratação, com emissão de gases de baixo peso molecular; 3) degradação de moléculas maiores, resultando na formação de carbono, gases condensáveis e não condensáveis; e 4) craqueamento de voláteis, gerando carbono e gases não condensáveis [23].

Most studies on the fast pyrolysis of biomass consider cellulose,

* Corresponding author.
E-mail address: cap@ufba.br (C.A.M. Pires).

https://doi.org/10.1016/j.jaap.2024.106919
Received 23 October 2024; Received in revised form 10 December 2024; Accepted 14 December 2024
Available online 16 December 2024

hemicellulose, and lignin as the main components responsible for bio-oil characteristics [1,24–28]. However, it is widely known that bio-oil quality is also related to the catalytic activity of inorganic species and the interference of extractives in the reactions (proteins, starch, waxes, fats, resins, tannic acid, pigments, sugars, pectin, saponins, essential oils, etc.), which varies according to their content in the biomass [29–34].

It is not easy to decouple the pyrolysis processes of different substances and study the corresponding product characteristics separately, which would help gain a more in-depth and comprehensive understanding of the pyrolysis properties of biomass extractives [35]. Due to the relatively low extractive content in biomass compared to the three main components, their influence on biomass pyrolysis is sometimes overlooked. However, they affect pyrolysis behavior and bio-oil properties, especially when pyrolyzing biomass rich in extractives, such as sisal residue, mahua seed, and banana leaves [2,36,37]. The way extractives influence the pyrolysis process of the three main components has been the subject of some researchers' studies [38–41]. When the extractive concentration is typically low (woody residues), extractives are considered to behave similarly to hemicellulose or lignin [42]. However, in non-woody agricultural residues, the extractive content can exceed 10 % [2], making their quantity and chemical composition crucial in determining the yields and characteristics of pyrolysis products [43].

The bio-oil obtained from the fast pyrolysis of biomass with high extractive content usually has two phases, which are formed due to the significant polarity, solubility, and density differences between lipophilic and hydrophilic extractives [44,45]. Lipophilic extractives are found in the upper layer of the bio-oil and have been periodically studied [46–49]. However, hydrophilic extractives, typically denser than lipophilic ones, have yet to be studied regarding the distribution of their chemical groups in bio-oil [33,34].

The state of the art in lignocellulosic biomass pyrolysis highlights extractives' critical role in modifying the resulting products' thermal and chemical properties. The presence of extractives facilitates the decomposition of cellulose, hemicellulose, and lignin, reducing the activation energy required and enhancing interactions among their volatile products [31,50,51]. Specifically, extractives catalyze acetic and formic acid formation while inhibiting cellulose levoglucosan production [31,50,52]. In hemicellulose, extractives exert a marginal influence on the devolatilization rate, contrasting with their more pronounced effects on cellulose and lignin, due to interactions with inorganic metals such as copper and manganese [23,51]. For lignin, extractives impact the generation of phenolic compounds and methanol, notably through the breakdown of methoxyl and methylene groups, which govern the distribution of gaseous and liquid products [31,38]. Thus, extractives influence the energy efficiency of pyrolysis and shape the final product composition, depending on their structure and chemical composition.

This article provides an overview of the state of the art on the influence of extractives on the decomposition of biomass macromolecules, which can play a crucial role in developing biorefinery projects. In this context, special attention was given to the characteristics of lignocellulosic biomass; biomass pyrolysis, fast pyrolysis of the individual macro-components of biomass (cellulose, hemicellulose, and lignin), and extractives; and lastly, state of the art on the influence of extractives on the devolatilization of macromolecules was discussed.

2. Lignocellulosic biomasses

Lignocellulosic biomass, derived from plant sources, is abundant and low-cost. It represents a renewable energy source that can yield equivalently 30–240 barrels of oil per hectare per year [52].

Regardless of the type, biomass contains three main components: cellulose, which is a porous structural material; hemicellulose, which acts as a matrix material; and lignin, which functions as a binder; in addition to extractives, which perform vital functions such as energy storage and external protection (Fig. 1).

The chemical structure and composition of biomass vary depending on its origin and type (Figs. 2 and 3). The cellulose, hemicellulose, and lignin in wood typically account for approximately 90 % of the total biomass, with the remaining 10 % corresponding to extractives, which are made up of molecules responsible for color, smell, taste, resistance to decay, and other properties (Fig. 2). However, extractives from leaves, grasses, straws, and parts of fruits can be found in significant amounts (sisal residue – 65 % [2]; mahua seed - 47 % [36]; corn cob – 27 % [54]). In addition to these, biomass also contains secondary inorganic components, such as sodium, potassium, calcium, silicon, phosphorus, and magnesium, which may exist as water-soluble salts, minerals, or organic compounds.

Biomass contains moisture in the form of hygroscopic or adhesion water (mainly in cellulose and hemicellulose), capillary water (present in the lumens of the biomass), and water vapor (in the gas phase) [32,61,62].

Fig. 3 shows moisture levels below 10 wt% (except for Eucalyptus chips), which is the maximum recommended content to produce a homogeneous bio-oil [1]. The volatility of biomass is mostly above 70 wt %, indicating that good yields of condensable and non-condensable gases are achievable, regardless of the type of residue, such as those from wood, leaves, straw, or parts of fruits. However, the high ash content of some biomass types, such as sisal residue (16.28 wt%), banana leaves (6.20 wt%), sugar cane bagasse (20.63 wt%), and rice straw (15.45 wt%), can have negative impacts on bio-oil production and quality because their alkali metals, like potassium and sodium, can cause secondary cracking of vapors [63].

2.1. Cellulose

Cellulose is the primary structural component of biomass, typically accounting for 20–60 % of the dry mass. It is a natural linear-chain polymer composed of D-glucose (pyranose) units linked by β-1,4-glycosidic bonds. The chains are held together by intra- and intermolecular hydrogen bonds, resulting in a highly crystalline structure with limited amorphous regions. This structure gives cellulose its remarkable mechanical strength and chemical stability. Cellulose microfibrils bundle together to form fibers, contributing to its robust characteristics. Additionally, cellulose can be considered an anhydro glucopyranose polysaccharide due to the loss of a water molecule during bond formation [32,53,64].

2.2. Hemicellulose

Hemicellulose is an important structural polymer that acts as a binding link between cellulose and lignin in the plant cell wall. It is present in biomass in amounts typically ranging from 10 wt% to 40 wt %, although it is rarely present in quantities higher than cellulose, as in some cases mentioned in Fig. 2. Its principal function is to provide structural support to the cell wall, contributing to the plant's integrity and stability [18,32]. Hemicellulose consists of a heterogeneous group of branched polysaccharides, including hexoses and pentoses, with monomers such as glucose, xylose, mannose, galactose, and arabinose. In addition, it contains glucuronic, galacturonic, and methylglucuronic acids. Unlike cellulose, hemicellulose has shorter chains and a more amorphous structure, with less physical strength due to its irregular composition and branched chains. Cellulose and hemicellulose form holocellulose, representing the total polysaccharide content [32,53,64,65].

2.3. Lignin

Lignin is a natural, amorphous macromolecule that acts as a binding agent between cells, providing rigidity to the cell walls due to its complex chemical composition. Cellulose microfibrils are coated with

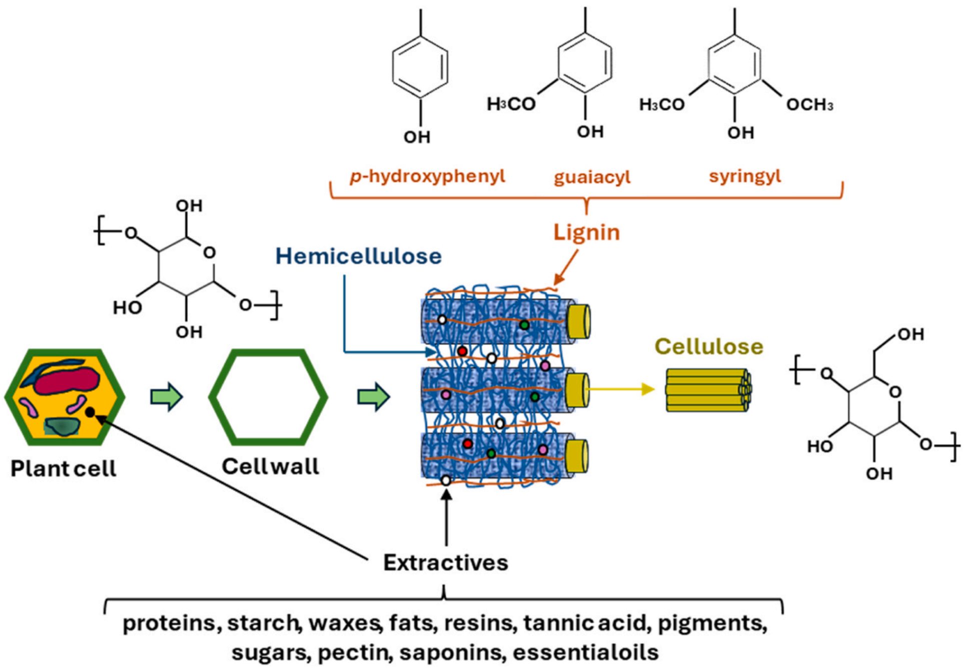

Fig. 1. – Simplified diagram of cell wall composition.

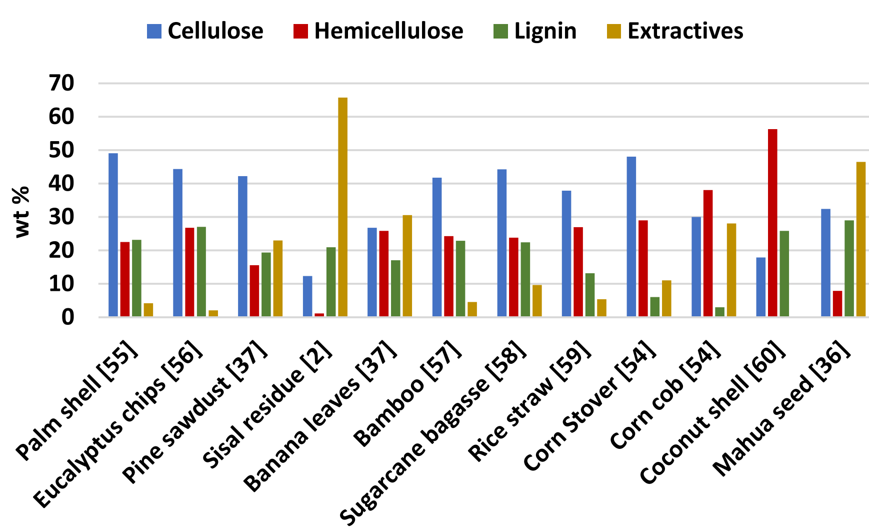

Fig. 2. Biochemical analysis of selected biomass (Dry-Ash-free basis).

hemicellulose in the cell walls, and the voids are filled with lignin. It is a three-dimensional aromatic phenolic polymer composed of phenylpropane units substituted with hydroxyl and methoxyl groups arranged randomly. These units, which include guaiacyl, syringyl, and p-hydroxyphenyl (Fig. 1), are linked by carbon-carbon (C-C) and carbon-oxygen (C-O-C) bonds, the quantity of which varies depending on the type of biomass, making lignin degradation more challenging [32, 64–66]. The percentage of lignin can vary from 23 % to 33 % in softwoods (pine and palm) and from 16 % to 25 % in hardwoods (eucalyptus) [53]; however, the variation in lignin content in other parts of the plant is much broader, as shown in Fig. 2.

2.4. Extractives

Extractives are organic compounds that play vital roles in plants, such as energy reserves and protection against microbes and insects [67]. They accumulate in the middle lamellae, inside cells, and in-plant canals (Fig. 1). Melzer et al. [46] point out that the presence of extractives significantly affects the physicochemical properties of solid biomass; a higher extractive content results in more carbon (C) and hydrogen (H), increasing the Higher Calorific Value (HHV). However, the HHV can be reduced by a high mineral content, as these contribute little to the energy released during biomass oxidation, especially in materials with up to 27 % ash [68].

Extractives vary widely in quantity, composition, structure, and biological functions, depending on the season, species, plant age, and region, such as those found in sugarcane bagasse (Fig. 4). This difference in extractive content can be justified by the diversity of genotypes used, as exemplified in the works of Masarin et al. [69], Philippini [70], Hoang et al. [71] and Kuchelmeister et al. [72].

They are distributed unevenly among plant organs, being more common in leaves and bark. Due to the great diversity of substances present in extracts, it is essential to identify the most common groups of molecules to classify and adequately represent different species [32].

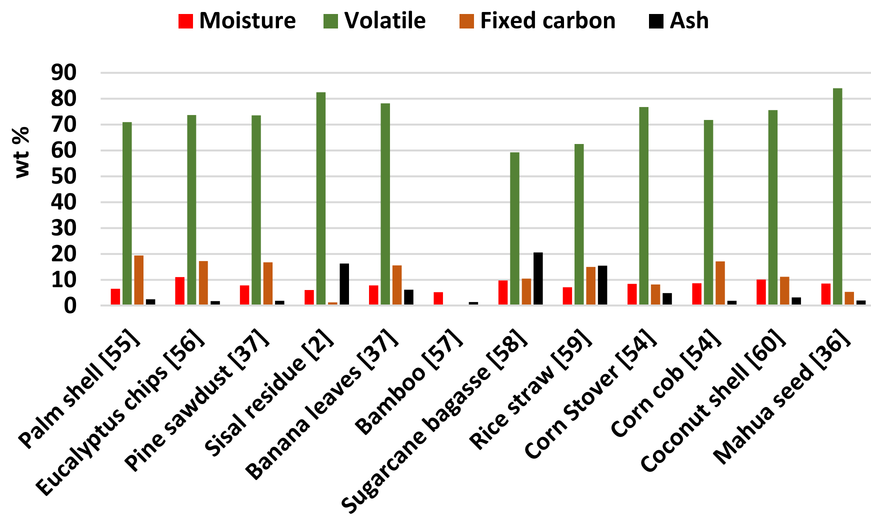

Fig. 3. – Proximate analysis of selected biomasses. In the case of bamboo, the work did not provide data on volatile and fixed carbon contents.

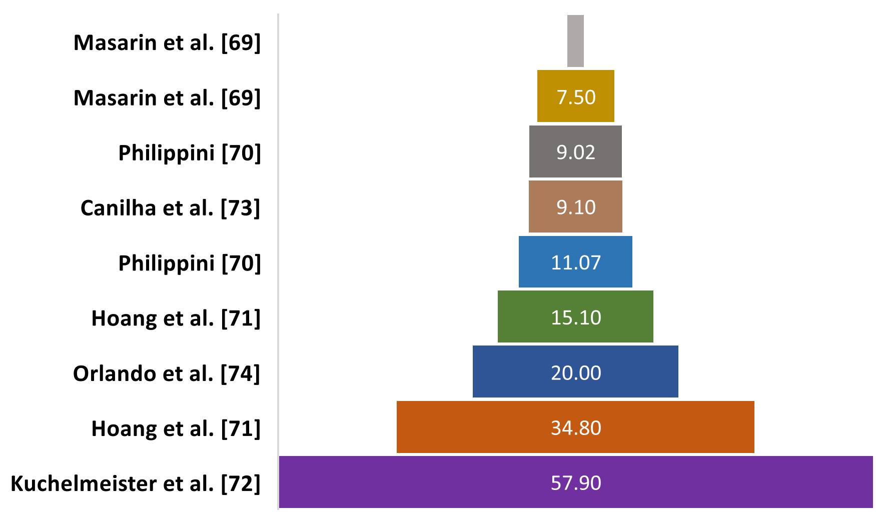

Fig. 4. Extractive contents for sugarcane bagasse.

Extractives contain a wide range of phenolic compounds, some of which are by-products of lignin biosynthesis. These compounds include steroids, essential oils, fats, glycosides, gums, mucilages, pectins, phenolics, proteins, resins, saponins, simple sugars, starches, terpenes, and waxes. They can be extracted using polar solvents, such as alcohol and water [71–74], or nonpolar solvents, such as benzene and toluene [61, 75]. Hydrophilic compounds, which dissolve well in highly polar solvents, are mainly phenolics, such as flavonoids and others. These compounds have antioxidant properties and are mainly found in leaves and bark. Hydrophobic compounds dissolve better in low-polar solvents, including terpenes and fatty acids, which can be free or esterified [76]. The content of water-soluble extractives also varies with the position from which the sample was taken from the plant; higher sections produce a greater quantity of extractives and a smaller quantity of cellulose [76].

The main aromatic compounds present in the extractives include vanillic and coniferyl alcohols, aldehydes such as vanillin and syringaldehyde, the ketone acetovanillone, and vanillic or syringic acids, present free or as products of lignin degradation [53,64].

3. Biomass pyrolysis

Fast pyrolysis is an endothermic process that requires between 207 and 434 kJ kg^{-1} of heat to decompose various agricultural biomasses and woods [20,21,53], resulting in the formation of liquid products (tars or bio-oil), solids (biochar), and low molecular weight gases. Typical yields are approximately 75 % for bio-oil, 12 % for biochar, and 13 % for gas. This process involves cracking, cyclization, aromatization, and dehydrogenation reactions, generating organic vapors that condense into bio-oil. The products formed depend on factors such as the biomass composition and the process conditions [20,53,77,78].

During the fast pyrolysis of biomass, the four main components undergo different reaction pathways, resulting in variations in the distribution and composition of the product, as shown in Fig. 5.

The physicochemical characteristics of bio-oil also depend on the type of reactor, which may influence the temperature profile of the reaction mixture. The decomposition temperature affects the conversion rate of the biomass components, influencing the amount of volatile products (gases and liquids) and biochar formation. However, differences in the characteristics of bio-oils caused by the type of reactor can be minimized if pyrolysis is rapid and the temperature profile is optimized. These characteristics were found in the works referenced in Figs. 6–8 involves micro-pyrolizers (fixed beds) and fluidized bed reactors.

The atomic ratios O/C (oxygen/carbon) and H/C (hydrogen/carbon) are crucial indicators of oil quality and are especially relevant for

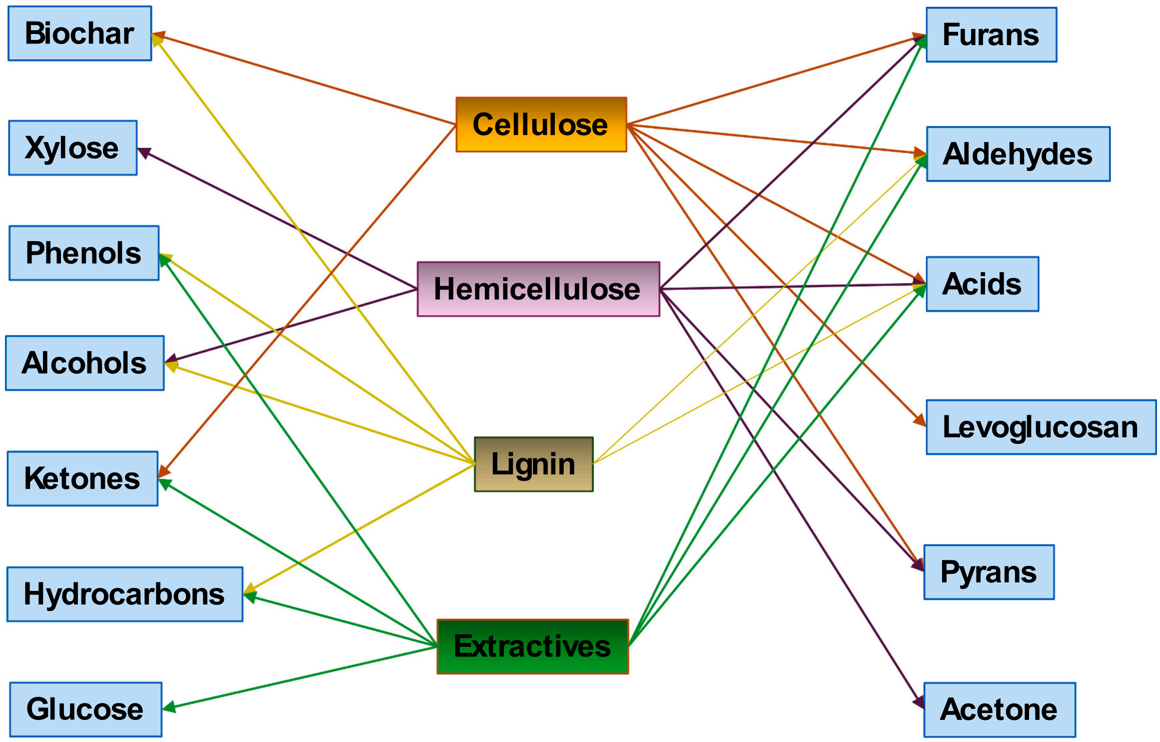

Fig. 5. Main products obtained from the thermal treatment of lignocellulosic biomass components [21,34,46].

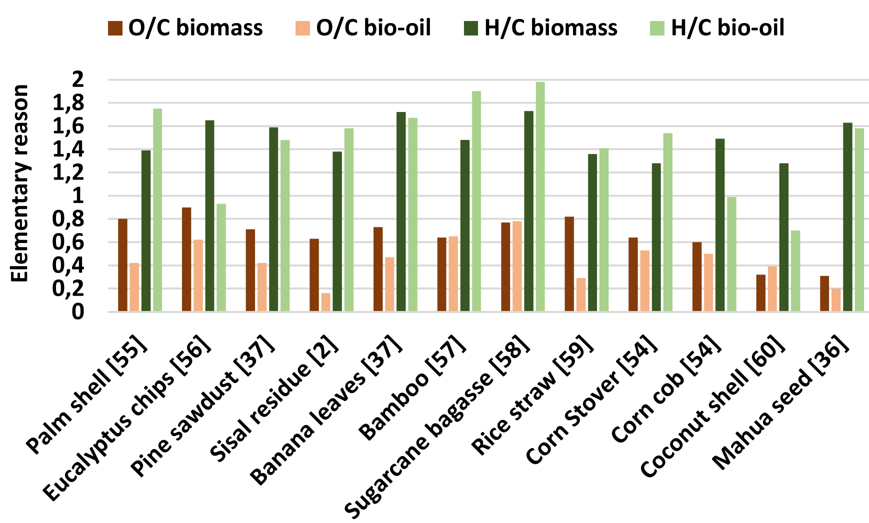

Fig. 6. – O/C and H/C ratio of bio-oils produced in fluidized [37,54,56 –59], and fixed [2,36,55,60] bed reactors at 500–575 °C of selected biomass.

monitoring processes such as hydrotreating and biodegradation [79]. Biomasses with high hydrogen and low oxygen content produce good-quality oils. Fig. 6 illustrates the O/C and H/C ratios of different types of biomass, such as wood, leaves, grasses, straws, and fruit parts, as well as their variations due to the conversion process into bio-oils. Generally, a reduction in the O/C ratios is observed during the thermoconversions of biomasses, probably due to the cracking of oxygenated molecules and the production of non-condensable molecules [36]. It is also noted that most of the O/C ratios of bio-oils are above 40 %, and their intensities vary within each type of biomass without a specific logic. However, bio-oils that stand out due to very low O/C ratios, such as sisal residue (0.16), mahua seed (0.20), and rice straw (0.29), are characterized from this point of view as the highest quality. Regarding the H/C ratio values, we observed that half of the biomasses presented in Fig. 6 had the ratio increased after pyrolysis, indicating that the loss of oxygen was caused preferentially by the production of CO and CO_2 instead of H_2O. However, most of the H/C values presented are close to the widely accepted values for petroleum (1.5—2.0).

The H/C ratio directly relates to the higher calorific value of the oils (HHV) [2]. However, when comparing Figs. 6 and 7, this relationship is not always observed, probably due to the water content of the bio-oils analyzed. The highest water contents identified for the biomasses in Fig. 7 were for sugar cane bagasse (40.21 wt%), pine sawdust (39.40 wt %), bamboo (15.82 wt%), corn cob (14.00 wt%), and palm shell (13.50 wt%), with corresponding HHVs of 18.5 MJ/kg (H/C=1.98), 23.8 MJ/kg (H/C=1.48), 18.3 MJ/kg (H/C=1.90), 26.2 MJ/kg (H/C=0.99) and 26.5 MJ/kg (H/C=1.75), respectively. Therefore, the theory would indicate sugar cane bagasse and bamboo as sources of bio-oils with the highest HHVs, but Fig. 7 does not confirm this trend due to their higher water contents [58]. However, the influence of

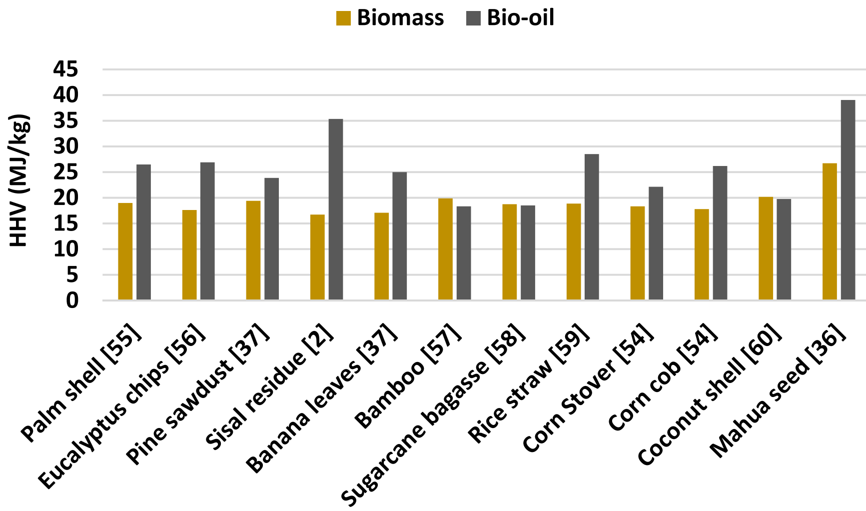

Fig. 7. High heating value (HHV) of bio-oils produced in fluidized [37,54,56–59] and fixed [2,36,55,60] bed reactors at 500–575 °C of selected biomass.

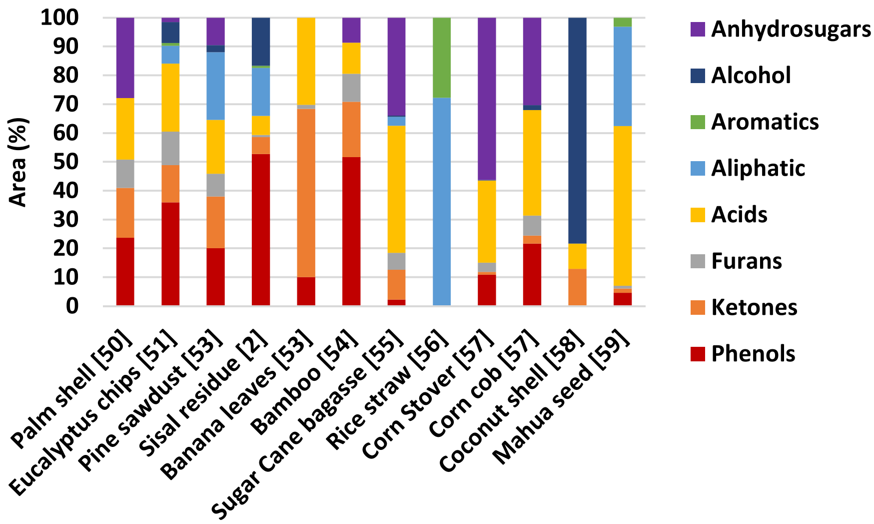

Fig. 8. Quantification of some bio-oil components produced in fluidized [37,54,56–59], and fixed [2,36,55,60] bed reactors at 500–575 °C of selected biomass.

measuring procedures and equipment in analyzing heating values cannot be despised. From a chemical point of view, a higher H/C ratio would indicate that the bio-oil is rich in saturated hydrocarbons, with paraffinic hydrocarbons generally having a higher HHV. In contrast, oils with a lower H/C ratio tend to be richer in aromatic or oxygenated compounds with a lower HHV. Higher amounts of heavy oxygenated compounds also produce higher HHV, indicating reduced deoxygenation [39].

The fast pyrolysis process involves a variety of reactions between substances present in the raw materials, intermediate substances, and final products [80]. Due to the complexity of describing each specific reaction and identifying all the substances involved, it is expected to generalize the reactions and group the substrates and products (Fig. 8). The complexity of biomass pyrolysis arises from the different forms of decomposition of its components, which have different reaction mechanisms and reaction rates influenced by the thermal processing conditions and the design of the reactors [40].

Lignin plays a crucial role in wood decomposition. Cellulose devolatilization does not reach its maximum rate until the β-O-4 bonds of lignin and the connections between hemicellulose and lignin are disrupted. Furthermore, removing lignin decreases the sample's thermal stability, an effect that is more pronounced than that observed with the removal of extractives [76].

The interaction between hemicellulose and lignin favors the production of lignin-derived phenols while hindering the formation of hydrocarbons, as observed by Kan et al. [81]. In contrast, the interaction between lignin and cellulose presents distinct effects. Lignin inhibits the polymerization of levoglucosan, a cellulose pyrolysis product, which reduces the formation of biochemical compounds. Furthermore, lignin stimulates the formation of low molecular weight products during cellulose pyrolysis [82]. Pyrolysis products derived from lignin, such as guaiacol or secondary char, are influenced by the presence of cellulose, and their production can be increased or reduced, depending on the conditions [83,84]. The distribution of tar and char is also significantly affected by the interaction between cellulose and lignin or inorganic salts [82]. The relative concentrations of lignin and cellulose in the biomass directly influence the calorific value and yield of the products [68].

Concerning the interaction between cellulose and hemicellulose, the impact on the formation and distribution of pyrolysis products is relatively minor, according to Kan et al. [81]. Cellulose, hemicellulose, and other nitrogen-free extracts, such as pectin, starch, and sugars, present similar behaviors and pyrolysis mechanisms [40].

The primary char formed during pyrolysis can act as a catalyst, promoting the cracking of primary organic vapors and forming secondary char, gas, and water [81]. However, this catalytic activity of primary coal tends to reduce bio-oil yields, which can be an important consideration in optimizing biomass thermal conversion processes [26,68].

The presence of extractives in biomass is crucial in the reactivity during pyrolytic conversion. They reduce the activation energy of woods such as pine and ash, as observed by Korus et al. [50]. This result is because extractives can degrade at low temperatures, forming a "film" that covers the surface of the reactant. This layer accelerates the reaction by promoting the transfer of condensation energy to the reactant and increasing collisions between molecules, which increases the pre-exponential factor [51,85,86].

Fast pyrolysis experiments show that increasing the extractive content in the initial biomass leads to higher liquid yields while the char and gas yields decrease [46]. For example, pyrolysis of extracted pine produces less acids and significantly more CO_2 and water than raw biomass pyrolysis [31]. Furthermore, the increase in extractives in the biomass results in higher percentages of carbon (C) and hydrogen (H), increasing the upper calorific value (HHV) of the gases due to the higher concentration of short-chain hydrocarbons. This increase in HHV has an exponential relationship with the extractive content, as also observed by Melzer et al. [46].

Extractives not only increase the calorific value of the biomass [87] but also facilitate the decomposition of lignin and promote the formation of phenolic homologs without causing significant thermal differences between the original biomass and the extracted residues [88]. In addition, they influence the density of the coal, making it difficult to drag it out of the reactor [46].

Another notable effect is the increase in pyrolysis oil yield, proportional to the extractive content in the biomass. A high extractive content allows the production of a two-phase bio-oil due to the lower polarity and oxygen content compared to lignocellulose compounds. These extractives accumulate in the upper phase due to their hydrophilic character [46].

3.1. Fast pyrolysis of pure cellulose

Cellulose can react by three distinct mechanisms: free radical, combined, and ionic. The glycosidic bonds in cellulose are unstable and can break in acidic environments or high temperatures [89]. During cellulose pyrolysis, the fundamental reaction occurs below 300 °C, involving decomposition and polymerization that form low molecular weight compounds, such as glycoaldehyde, furan, hydroxyacetaldehyde, formic acid, H_2O, and CO_2 [90]. The primary products of pyrolysis result from breaking glycosidic bonds and dehydration reactions, and levoglucosan (anhydrous sugarís) is an example of the products formed [89]. At temperatures above 300 °C, levoglucosan polymerizes, forming compounds with active functional groups such as –COOH, –COO–, –C═O, –OH, and C–O–C. Cyclization reactions also occur, producing stable oxygenated compounds such as furfural, 5-hydroxymethylfurfural, and pyran [91]. However, levoglucosan yield in cellulose pyrolysis is limited by the competition between levoglucosan production and its degradation and other reactions that produce light oxygenates and non-condensable gases [92].

During cellulose pyrolysis, secondary reactions are almost inevitable and can increase carbonization. The formation of oligomers from intermediates contributes to the undesirable conversion of cellulose to charcoal. However, compounds in charcoal by-products can undergo decarboxylation and decarbonylation reactions, which broaden the aromatic structure of biochar and release gaseous products [28,89].

3.2. Fast pyrolysis of pure hemicellulose

Studying hemicellulose pyrolysis is challenging due to the difficulty of separating it from the biomass without causing degradation [89]. Therefore, it is common to use xylan, the most abundant sugar in hemicellulose, to understand its behavior in pyrolysis. As in cellulose pyrolysis, hemicellulose pyrolysis produces light oxygenates such as water, acetone, methanol, acetic or formic acid, furfural, and anhydro xylopyranose, which can be formed competitively or consequentially. A significant difference is that hemicellulose pyrolysis can produce xylose after breaking the glycosidic bond, while cellulose pyrolysis does not generate glucose, possibly due to the preference for forming levoglucosan or the lower production of water. The decomposition and polymerization reactions occur below 300 °C, forming compounds such as glycolaldehyde, furan, acetaldehyde, formic acid, water, and CO_2 [90]. Cleavage of the glycosidic bond also occurs, forming oligosaccharides, which decompose into xylose, releasing water and CO_2. At higher temperatures, xylose undergoes conversion, producing compounds such as acetic acid, formic acid, furfural, and furan, through cracking and rearrangement of the depolymerized molecule.

In addition, the charcoal produced contains active functional groups such as C═O, COOH, –OH, and C–O and ring structures containing oxygen. Pyrolysis polymerization reactions form these compounds intermediates, releasing water, CO_2, and CO [93,94].

Wang et al. [95] compared the pyrolysis of hemicelluloses from hardwood and softwood, observing that hemicellulose from softwood has a higher activation energy and produces mainly 5-hydroxymethylfurfural and anhydrous sugars. In contrast, hemicellulose from hardwood generates mainly furfural and acids. Yang et al. [96] used TG-FTIR, Py-GC/MS (Pyrolysis Gas Chromatography Mass Spectrometry) techniques, and DFT (Density Functional Theory) calculations to investigate the detailed pyrolysis mechanism of hemicellulose. They found that the process involves cleavage of the primary glycosidic bond, followed by ring-opening, retro-aldol, retro-Diels-Alder, dehydration, and enol-keto tautomerization reactions, forming low molecular weight oxygenates.

3.3. Fast pyrolysis of pure lignin

Lignin is notoriously challenging to decompose and undergoes three consecutive stages during thermal degradation. First, water evaporates; then, primary volatiles are formed; and finally, small molecular gases are released between 160°C and 900°C. The main volatile compounds released are phenolics, alcohols, aldehydes, and acids, and the formation of gases such as CO, CO_2, and CH_4 [97,98].

Lignin is the main source of phenolic compounds in the liquid pyrolysis product, in addition to contributing oxygen-rich species (such as OCH_3), some hydrocarbons, CO, and CO_2 [78]. Lignin pyrolysis begins with the dehydration and decomposition of ether bonds, such as β–O–4, between 250 °C and 350 °C, followed by the polymerization of the products formed. The cleavage of aliphatic side chains begins around 300 °C, while the cleavage of the more stable C–C bonds between lignin monomers occurs at around 400 °C [99–101].

In the pyrolysis of softwood and hardwood lignin, the distribution of products varies with the reaction temperature, and most degradation compounds present yields below 1 % [102]. The highest yields observed were 5-hydroxy vanillin (4.29 %) in softwood lignin and 2-methoxy-4--vinylphenol (4.15 %) in hardwood lignin. Softwood lignin generated a higher carbonization index due to the more significant presence of C–C bonds. Phenolic compounds, such as phenol, 2-methoxy-4-vinylphenol, 4-vinylphenol, and 2,6-dimethoxyphenol, are often significant products [103]. These monomers tend to polymerize into oligomeric species during condensation, a process that can accelerate the presence of acetic acid.

3.4. Fast pyrolysis of pure extractives

Extractives can be divided into hydrophilic, soluble in polar solvents, and lipophilic, soluble in nonpolar solvents. Hydrophilic extractives predominantly comprise phenolic compounds, including lignans and tannins [104,105], which can be further classified into flavonoids and nonflavonoids [32,104]. On the other hand, lipophilic extractives mainly comprise fatty acids, alcohols, phytosterols, hydrocarbons, and triterpenes [75].

Few authors have studied the pyrolysis of biomass extractives, of which a small portion has been dedicated to studying the production of chemical species. Sampaio et al. [34] studied the micro pyrolysis of hydrophilic extractives (extracted with water and ethanol) from sugarcane bagasse (grass), sisal residue (leaf), and eucalyptus bark residue (wood). Fig. 9 shows a diversity of contents and chemical groups of the extractives and products of their pyrolysis.

The extractives from sugarcane bagasse, sisal residue, and eucalyptus bark bagasse are composed mainly of saccharides (89.14 %, 21.82 %, and 53.10 %, respectively), acids (2.10 %, 41.78 %, and 15.27 %, respectively), alcohols (1.08, 32.05 and 2.41 %, respectively), aromatics (6.77 %, 0.00 %, and 6.88 %, respectively) and steroids (0.98 %, 0.00 %, and 7.77 %, respectively). The pyrolysis of extractives from sugarcane bagasse, sisal residue, and eucalyptus bark bagasse produced new groups, with emphasis on non-condensable gases (24.43 %, 52.59 %, and 75.31 %, respectively), phenols (18.05 %, 16.07 % and 2.21 %, respectively), alkanes (4.59 %, 0.00 % and 10.28 %, respectively) and furans (19.41 %, 0.90 % and 0.00 %, respectively). In addition to these, some groups produced by the pyrolysis of extractives had percentages between 5 % and 10 %, such as nitrogenous and acids from sugarcane (8.05 % and 5.07 %, respectively) and aldehydes from sisal residue (5.19 %). Pyrolysis of saccharides produced non-condensable gases (CO_2), phenols, and furans. In the pyrolysis of sugarcane bagasse, ~27 % of saccharides produced species from other chemical groups such as alkanes, alcohols, acids, aldehydes, ketones, esters, and aromatics. However, saccharide pyrolysis contributed only a portion of the production of non-condensable gases, phenolics, and furans, since it was consumed entirely for sisal and eucalyptus residue.

Pyrolysis of extractive acids contributed to the formation of alcohols, aldehydes, and ketones. Substances such as furans and ketones indicate the thermal decomposition of water-soluble carbohydrates and represent the main products resulting from saccharide pyrolysis [35,106]. Some of the alcohols in the extractives were probably transformed into aldehydes and ketones by acid-mediated oxidation, while the carboxylic acids may have undergone decarboxylation to form CO_2 and saccharides in aromatics [34].

Qiao et al. [40] performed the pyrolysis of eucalyptus residue lipids, which generated small oxygenated compounds, forming volatiles by cleavage of C–C and C–H bonds. Por outro lado, a pirólise de proteínas produziu amônia, que foi transformada em nitrilas e amidas pela combinação com outros intermediários, tornando-se os principais produtos voláteis.

Peng et al. [35] separated tobacco extractives with two polar solvents, associating the pyrolysis products with temperature stages. The changes indicate that different compounds are formed or decomposed at different temperature ranges, reflecting the reactions' complexity. The main components of the original extractives were nicotine, with 67.74 % (water) and 92.32 % (dichloromethane), followed by ketone, with 27.72 % (water) and 2.25 % (dichloromethane), and aromatic with 5.34 % (dichloromethane). The molecular structure of nicotine is relatively complex, containing pyridine and pyrrolidine rings. During pyrolysis, these structures break and rearrange, resulting in a mixture of volatile and non-volatile products. Nicotine and ketone were initially transformed into alcohols (10.73 %), furans (22.00 %), and aliphatics (11.36 %); in a second moment, the remaining nicotine was almost all degraded (0.75 %), as well as alcohol (0.0 %) and furans (5.55 %), producing the first significant content of phenols (48.91 %); then, a little phenol (11.92 %) and most of the aliphatic (12.09 %) were transformed into 19.84 % of furans and 5.42 % of aromatics; at temperatures above 477.9 °C, 32.09 % aromatics, 34.32 % pyrans, and 6.78 % aliphatics were produced, probably due to the consumption of 21.72 % phenols, 22.79 % ketones, and 23.26 % furans.

According to Tian et al. [107], thermal degradation of amino acids between 300 and 500 °C results in the formation of amines, which, through secondary reactions such as dehydrogenation and polymerization, can generate nitriles and N-heterocycles as the temperature increases. Between 600 and 700 °C, fatty acids from lipid decomposition react with ammonia released from proteins, forming long-chain amides, which release nitriles such as acetonitrile and 2-propene-nitrile upon dehydration. Subsequent cyclization, condensation, and dehydration reactions between amines form N-heterocycles, such as pyridine, pyrrole, and indole. In addition, the interaction between nitrogenous compounds and glucose depolymerization products, such as furan and pyran, can release N-heterocycles through the Maillard reaction. Reducing ketone compounds formed by Amadori intermediates react with amino acids, undergoing degradation, cyclization, and rearrangement processes to form N-heterocycles and other nitrogenous compounds [40].

4. Influence of extractives on the decomposition of macromolecules

Section 3 clearly shows the intersections between pyrolysis products from cellulose, hemicellulose, lignin, and extractives, but in an isolated manner. When the pyrolysis of lignocellulosic biomass is analyzed, each of these components has a distinct behavior, and the presence of extractives can modify their influence on the final products.

The activation energies of the biomasses, before and after extraction, indicate that the absence of extractives requires greater activation energy for pyrolysis. Therefore, the presence of extractives facilitates the decomposition of the structural components of the biomass [31,50,51]. Poletto et al. [51] demonstrated that higher levels of extractives with

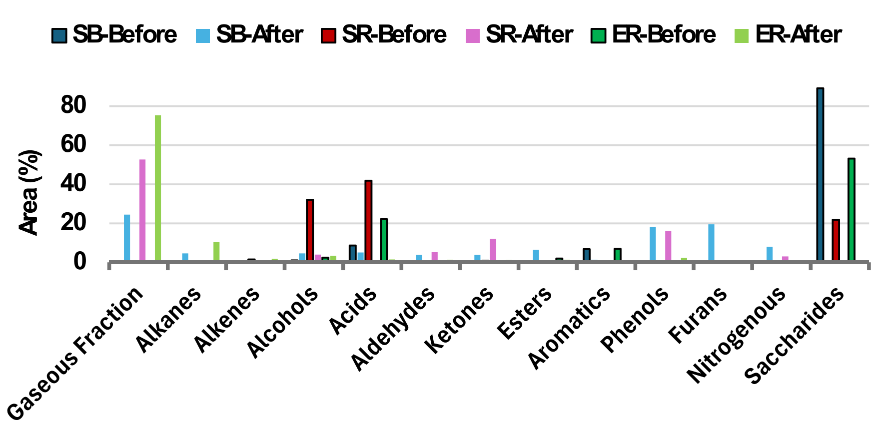

Fig. 9. – Identification of product composition of water/alcohol-soluble extractives of sugarcane bagasse (SB), sisal residue (SR), and eucalyptus bark residue (ER). (Sampaio et al. [34]).

affinity for benzene/ethanol accelerate degradation and reduce the thermal stability of the biomass. The extractives decompose at low temperatures, and the contact of the products with the main components of the biomass intensifies their reactions. This behavior effect occurs due to the transfer of energy released by the condensation of extractives to the reactants, increasing collisions between molecules [82]. Furthermore, extractives significantly influence the devolatilization rate more than cellulose, lignin, and nutrient contents [23]. Phenolic extractives, in particular, are one of the main components that affect biomass degradation, reducing degradation mechanisms [31].

The presence of extractives in pine wood favors the formation of charcoal over liquid products, making it less microporous and, to some extent, less acidic. This phenomenon is explained by the increased residence time of tar in the extractive-rich material and by the catalytic action of inorganic species, which can be removed by polar solvents during extraction [50].

4.1. Influence of extractives on the decomposition of cellulose

Cellulose is composed of repeating glucose units, which can be broken down at high temperatures, contributing mainly to the formation of furans and levoglucosan [38]. Although glycoaldehydes, hydroxy acetaldehydes, acids, H_2O, and CO_2 are formed from secondary reactions [89].

Some studies clarify that extractives inhibit the formation of levoglucosan and increase the formation of acetic and formic acid [31,50,108]. β-scission reactions within the polymer chain and successive molecular and radical reactions were responsible for the formation of acetic acid. Additionally, extractives catalyze the formation of water and CO_2 from the secondary cracking of levoglucosan. The presence of metal salts in biomass reduces the release of gases from small molecules, as it inhibits the breakdown of levoglucosan, blocking the free junctions of the cellulose chains. On the other hand, this increases the formation of acetic acid since the polymer chains break internally. [52].

The presence of extractives reduces the cellulose devolatilization rate, with a more pronounced effect than on hemicellulose [23]. The maximum rate of devolatilization of extractives occurs at approximately 395 °C, slightly higher than the maximum degradation temperature of cellulose, which ranges from 342 °C to 379 °C. This behavior suggests a greater association of extractives with cellulose than with hemicellulose. This result contradicts Poletto et al. [51], who state that extractives intensify the devolatilization rate. This contradiction may occur since some inorganic metals in biomass can inhibit devolatilization. Copper (Cu) and manganese (Mn) are examples of metals that, together with an increase in extractives, negatively impact the cellulose devolatilization rate, decreasing it. Such a relationship suggests a strong association between extractives and cellulose, possibly with lignin, resulting in lower cellulose devolatilization [23].

4.2. Influence of extractives on the decomposition of hemicellulose

During pyrolysis, hemicellulose decomposes into a complex mixture of products, such as non-condensable gases (CO_2, CO, CH_4, and H_2) and liquids (furfural, aldehydes, carboxylic acids, ketones, and anhydrosugars). Acetic acid is one of the main liquid products generated [38]. The devolatilization rate of hemicellulose is only slightly influenced by the chemical properties, the amount of cellulose or lignin, or the extractive content. The value of the devolatilization rate of hemicellulose was positively affected only by the content of nitrogenous species; therefore, this compound is the element causing the differences with cellulose. This effect may have been caused by the greater volatility of nitrogen relative to other nutrients [23].

4.3. Influence of extractives on the decomposition of lignin

Lignin is a complex molecule formed by phenylpropane units, making its pyrolysis different from cellulose and hemicellulose, which are carbohydrates. Several phenolic compounds, such as guaiacol, syringol, and cresol, are produced during lignin pyrolysis. Acids such as acetic and formic acid are also generated in smaller quantities than cellulose and hemicellulose. In addition, small amounts of benzene, toluene, and xylene can be formed [38]. Lignin pyrolysis releases gases such as CO_2, CO, CH_4 [31,38]. Therefore, the influence of extractives on lignin pyrolysis is complex and depends on the specific composition of the extractives present and the pyrolysis conditions (such as temperature, heating rate, and atmosphere). For Guo et al. [31], the influence of extractives on the distribution of the biomass pyrolysis product is caused by the different types and content of lignin units (guaiacyl and syringyl units). The higher content of methoxyl groups these units during the pyrolysis of extractive increased the formation of methanol. In contrast, breaking methoxyl and methylene groups is crucial for methane formation. Hence, the content of the methoxyl group results in the difference in product distribution of extractive pyrolysis.

4.4. Distribution of pyrolysis products influenced by extractives

The study of the influence of extractives on biomass pyrolysis is scarce, especially when the focus is on the distribution of biomass pyrolysis products (Table 1).

In the work of Sampaio et al. [34], the main volatile fractions obtained from the original biomasses were CO_2 (for all biomasses), aldehydes (sugar cane and eucalyptus residue), saccharides (sugar cane and eucalyptus residue), alkanes and alkenes (sisal residue) and furans (sugar cane).

The pyrolysis process is so complex that the production of non-condensable gases may have come from the decomposition of holocellulose, lignin, and extractives [34,38,64]. Sampaio et al. [34] attributed the high pyrolysis temperature as the most likely cause for the high CO_2 production, which can be justified by the significant cracking of macromolecules [110].

The influence of extractives on CO_2 production was discussed by Guo et al. [31] and Wan et al. [39], who considered the absence of extractives as a catalyst for CO_2 formation, while their presence increases the formation of acidic compounds and water. The inversely proportional relationship between CO_2 and extractives found by Wan et al. [39] applied only to sisal residue [34] and tobacco [82], which presented the highest extractive contents (44.3 % and 47.57 %, respectively). Pagano et al. [109] also considered that extractives inhibited CO_2 formation from wheat straw pyrolysis but enhanced CO production. These results suggest that alkali and alkaline earth elements in ash directly promote decarboxylation reactions about decarbonylation [39]. Additionally, Guo et al. [82] stated that CO_2 generation was inhibited with the increase in extractives, producing furans, phenols, and toluene in different degrees due to the interaction between the components.

The directly proportional relationship between acid production and the existence of extractives indicated by Guo et al. [31] does not apply to all biomasses. In the case of Sampaio et al. [34], the relationship was only valid for sugarcane bagasse but did not apply to sisal and eucalyptus residue, as occurred in Araújo et al. [33] when pectin was removed from sisal residue.

Yang et al. [111] reported that CH4 was mainly derived from lignin and cellulose. Shi and Wang [112] observed that, in addition to lignin and cellulose, "immature" charcoal formed by devolatization of hemicellulose also acted as an important precursor of CH_4. However, Wan et al. [39] suggested that neutral extractives from peanut straw may have contributed significantly to the formation of CH_4 since its content was one of the highest (1.12 %), even with intermediate cellulose (26.8 %) and low lignin (7.6 %) contents compared to other biomasses studied. Acetic acid is one of the main acids produced from the degradation of hemicellulose and, to a lesser extent, from the direct depolymerization and fragmentation of cellulose in the presence of potassium [39,109]. Acetic acid is formed by eliminating acetyl groups attached

Table 1
Main references on the effect of extractives on the composition of bio-oil.

Reference	Biomass	Extractive	Extractives Pyrolysis	Solvent	Reactor	Temperature
[31]	Mongolian pine Manchurian ash	8.7 % 4.3 %	H_2O, CO, CO_2, propanal, ácido fórmico, ácido acético, methanol and etanol H_2O, CO, CO_2, methanol, methane, aldehydes, and phenolics	Ethanol	TG-FTIR	25–800 °C
[33]	Resíduo de sisal	65 %	CO, CO_2, aliphatic, acid, saccharides, esters, aromatics, furans	Water/Ethanol	fixed bed	450 °C
[34]	Sugarcane bagasse Sisal residue Eucalyptus residue	10.6 % 44.3 % 3.1 %	CO, CO_2, alkanes, alcohols, acids, aldehydes, ketones, esters, phenols, furans, nitrogenous, saccharides CO, CO_2, alkenes, alkenes, alcohols, acids, aldehydes, ketones, esters, aromatics, furans, nitrogenous, saccharides CO, CO_2, alkenes, alkenes, alcohols, acids, aldehydes, ketones, esters, phenols, furans, nitrogenous, saccharides	Water/Ethanol	fixed bed	550 °C
[39]	Peanut straw Cotton straw Sorghum stalk Corn stal Reed	47.4 % 14.9 % 16.8 % 17.4 % 15.6 %	C2-C5, C6-C30, alicyclic, saccharides, furans, phenols, fatty, aromatic C2-C5, C6-C30, alicyclic, saccharides, furans, guaiacols, fatty, aromatic C2-C5, C6-C30, alicyclic, saccharides, furans, guaiacols, phenols, fatty, aromatic C2-C5, C6-C30, alicyclic, saccharides, furans, guaiacols, phenols, fatty, aromatic C2-C5, C6-C30, alicyclic, saccharides, furans, guaiacols, phenols, fatty, aromatic	Sodium lauryl sulfate (neutral detergent)	fixed bed	25–700 °C
[46]	Cashew shell Jatropha Shea	15.3 % 12.3 % 14.9 %	Acids, phenols, aromatic, PAH's Acids, aldehydes, ketones, phenols, aromatic, guaiacols, PAH's Acids, aldehydes, ketones, phenols, aromatic, PAH's	Oil extraction by pressing/ Hexane	fixed bed	500 °C
[82]	Tobacco	47.57 % 14.14 %	Furans, ketones, hydrocarbon, phenols, nicotine, pyridines Furans, ketones, hydrocarbon, phenols, nicotine, pyridines	Water/Ethanol	fixed bed	420 °C
[109]	Wheat straw	10.8 %	Ketones and ethers, acids, amines, furans, oxygenated aromatics, sugars	Water, NH3, Acetic acid	fixed bed	550 e 450 °C

initially to the xylulose unit [113]. However, due to the minor variations in acetic acid values found between the original and extracted biomass in Sampaio et al. [34], it was suggested that the acetyl groups have their origins linked to the xylulose unit and extractives.

Hydrocarbons can be formed by the decomposition of hemicellulose [114], lignin [115], and extractives [75]. Wan et al. [39] vehemently stated that the neutral extractives of peanut straw were suitable precursors of aliphatic hydrocarbons. Additionally, in the pyrolysis of the extracted sisal residue, alkanes were reduced by 95.3 % and alkenes by 45 %, indicating that extractives enhance the formation of hydrocarbons from the original biomass [34]. Furthermore, higher extractive contents should produce higher hydrocarbon content. This statement is explicit in Sampaio et al. [34] when verifying that the pyrolysis of the original biomass formed hydrocarbons (sisal residue (46.3 %) > sugar cane (0.4 %) > eucalyptus residue (0.0 %)), following the same trend as the extractive contents (sisal residue (44.3 %) > sugar cane (10.6 %) > eucalyptus residue (3.1 %)).

Aldehydes are produced by the pyrolysis of biomass, mainly cellulose and hemicellulose [116]. The extractives enhanced aldehyde production by holocellulose, forming a large amount of this compound from sugarcane bagasse and eucalyptus residue (~23 %). The effect was the opposite for sisal residue since the extractives negatively interfered with aldehyde production from the original biomass [34]. This effect may have occurred due to the transformation of aldehydes into short-chain hydrocarbons promoted by the high extractive content of sisal residue.

The extractives inhibited ketones, esters, furans, nitrogenous compounds, and saccharides from sisal residue but enhanced the production of furans from sugarcane bagasse and nitrogenous compounds from eucalyptus residue [34]. The main source of ketones is holocellulose [117], and of furans is cellulose [118], showing the interference of extractives in these macromolecules. Pagano et al. [109] reached the same conclusions for ketones, ethers, and sugars produced in the pyrolysis of wheat straw washed with acetic acid; however, the authors considered that furans were increased.

Unlike short-chain aliphatic oxygenates, long-chain aliphatic oxygenates can be products originating mainly from neutral extractives rather than holocellulose [39]. Lignin is the main source of phenolics and although the lignin contents were similar (eucalyptus residue (27.9 %), sugarcane bagasse (26.3 %) and sisal residue (23.5 %)), the original biomasses presented low phenolic contents, but with significant differences for sugarcane bagasse (4.9 %), eucalyptus residue (3.0 %) and sisal residue (2.0 %) [34]. However, the influence of extractives on the thermal decomposition of lignin appears to be complex. In the case of sisal residue, the extractives did not influence the phenolics produced by the original biomass. However, they were slightly enhanced by the extractives from sugarcane bagasse (4.0 % for the extracted biomass) and significantly inhibited by the extractives from eucalyptus residue (7.2 % for the extracted biomass).

Nitrogen compounds are derived from degrading nutrients and pigments (proteins, vitamins, chlorophyll, etc. [39,110]. Proteins are formed by amino acids, which can be polar and nonpolar. In Sampaio et al. [34], the extractives were obtained by polar solvents, so a portion of the protein was part of the extracted biomass and produced nitrogenous species. Phospholipids and glycolipids are extractives that have nitrogen in their compositions and, when undergoing pyrolysis, can form glycerol. The addition of glycerol to the process can help in the devolatilization of hemicellulose and favor the formation of ketones, aldehydes, hydrocarbons, and carboxylic acids, in addition to helping to inhibit the production of aromatic hydrocarbons at high temperatures [114].

According to Wan et al. [39], long-chain fatty acids and long-chain aliphatic hydrocarbons can be derived from lipid constituents of neutral extractives in corn and sorghum stems. During pyrolysis, fatty acids generated from lipid breakdown interact with ammonia (NH_3), released from the pyrolysis of proteins, to produce long-chain amides. Subsequent dehydration of these amides generates significant quantities of nitriles [40]. Che et al. [119] found that nitrogenous tar contained heterocycles and amines when the bio-oil was produced without lipids in biomass. However, the presence of lipids significantly influenced protein pyrolysis, leading to the formation of amides and nitriles. Notably, lipids played a key role, producing 34.16 % amides from glutamic acid, compared to 9.17 % from histidine and 4.25 % from tyrosine.

According to Sampaio et al. [34] and Pagano et al. [109], extractives inhibit the production of saccharides formed from the degradation of holocellulose. The production of saccharides is directly proportional to the holocellulose content of the biomass, and there are no restrictions regarding this rule to date. The holocellulose contents of sugarcane bagasse, sisal residue, and eucalyptus residue were 52.2 %, 18.2 %, and 60.8 %, respectively, and the saccharide contents related to the pyrolysis of the original biomass and those extracted from sugarcane bagasse, sisal residue, and eucalyptus residue were 10.5 % and 18.9 %, ~0.0 % and 1.8 %, 16.6 % and 18.8 %, respectively [30]. Additionally, data from Pagano et al. [109] showed a significant increase in these compounds when wheat straw was washed with acetic acid (~10 %), ammonia (~4.8 %) or water (~3.7 %). In the presence of K and Mg in biomass ash, they can be converted into light-oxygenated species such as ketones and esters [109].

5. Discussion

Extractives, the non-structural components of biomass, play a critical and multifaceted role in pyrolysis, influencing thermal reactions and final product outcomes. One of their most notable characteristics is their ability to reduce the activation energy of the process, thereby accelerating thermal degradation reactions. This property positions extractives as natural catalysts capable of enhancing liquid product yields. However, the impacts of these compounds vary significantly across different biomass types, rendering their role more complex and less predictable.

Cellulose and hemicellulose, for instance, exhibit greater sensitivity to extractives, whereas lignin, known for its thermal resistance, experiences facilitated decomposition in their presence. Nonetheless, the impact of extractives on the formation of volatile products, such as acids, CO_2, and hydrocarbons, and the inhibition of desirable compounds like levoglucosan, raises questions about the selective control of their interactions. For example, sisal residues, characterized by a high extractive content, produce higher hydrocarbons. However, this observation requires robust experimental validation and reproducibility across other biomass types.

Extractives also directly affect bio-oil quality. Their presence tends to result in biphasic oils and increases the concentrations of less polar, lower-oxygen compounds—desirable characteristics for energy applications. However, the increase in less oxygenated compounds may also be associated with the formation of undesirable by-products or the need for additional stabilization treatments, impacting the economic feasibility of the process. In the case of charcoal, extractives lead to products with lower microporosity and higher acidity, limiting their applicability in adsorption-demanding uses, such as filters and catalysts.

The diverse effects of extractives across different biomasses underscore the need for more detailed characterization and mechanistic understanding of their interactions. Despite progress in identifying the general influences of extractives, the detailed mechanisms remain largely unknown. This gap limits the predictability of pyrolysis behavior and hinders the ability to adapt processes to specific feedstocks and target products.

Future perspectives should include integrating advanced experimental approaches with computational modeling to unravel biomass's chemical and thermal interactions between extractives and structural polymers. Moreover, strategies to manipulate the content and type of extractives could open new avenues for optimizing pyrolysis, tailoring it to maximize specific yields, improve product quality, and meet sustainability requirements. Therefore, a detailed understanding of extractives is essential for transforming challenges into opportunities in bioenergy and sustainable chemistry.

Conclusion

Extractives play a crucial and multifaceted role in pyrolysis, influencing both thermal reactions and the quality of the final products. While they can reduce activation energy and act as natural catalysts, their impacts vary widely across different biomasses, making their behavior difficult to predict. They significantly affect the formation of volatile compounds, the quality of bio-oil, and the properties of charcoal, highlighting the importance of more precise control over their interactions.

Despite progress in understanding their general effects, the detailed mechanisms of these interactions remain unclear, limiting the optimization of processes for different feedstocks and products. Future research should combine advanced experimental techniques and computational modeling to uncover these mechanisms, enabling the fine-tuning of pyrolysis processes. This understanding is essential for maximizing efficiency, improving product quality, and promoting sustainable solutions in bioenergy and renewable chemistry.

CRediT authorship contribution statement

Carlos Augusto de Moraes Pires: Writing – review & editing, Writing – original draft, Supervision, Formal analysis, Conceptualization. **Sirlene B. Lima:** Writing – review & editing, Methodology. **Thamyris Q. S. Sampaio:** Writing – original draft, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to acknowledge the support of the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) - Finance Code 001.

Data Availability

No data was used for the research described in the article.

References

[1] A.V. Bridgwater, Review of fast pyrolysis of biomass and product upgrading, Biomass-.-. Bioenergy 38 (2012) 68–98, https://doi.org/10.1016/j.biombioe.2011.01.048.
[2] T.A. Jambeiro, M.F.S. Silva, L.G.G. Pereira, D.Da.S. Vasconcelos, G.B. Silva, M. B. Figueirêdo, S.B. Lima, C.A.M. Pires, Fast pyrolysis of sisal residue in a pilot fluidized bed reactor, Energy Fuels v. 32 (n. 9) (2018) 9478–9492, https://doi.org/10.1021/acs.energyfuels.8b01718.
[3] L.G.G. Pereira, H.J. Heeres, S.B. Lima, C.A.M. Pires, Development of process-product relations for the pyrolysis of sisal residue, J. Anal. Appl. Pyrolysis 165 (2022) 105583, https://doi.org/10.1016/j.jaap.2022.105583.
[4] A.V. Bridgwater, D. Meier, D. Radlein, An overview of fast pyrolysis of biomass, Org. Geochem. 30 (1999) 1479–1493, https://doi.org/10.1016/S0146-6380(99)00120-5.
[5] Oasmaa, A., Testing standard methods. Fast pyrolysis of biomass: A handbook, 1999. 1: p. 69-74.
[6] Q. Zhang, et al., Review of biomass pyrolysis oil properties and upgrading research, Energy Convers. Manag. 48 (1) (2007) 87–92, https://doi.org/10.1016/j.enconman.2006.05.010.
[7] L. Qiang, L. Wen-Zhi, Z. Xi-Feng, Overview of fuel properties of biomass fast pyrolysis oils, Energy Convers. Manag. 50 (5) (2009) 1376–1383, https://doi.org/10.1016/j.enconman.2009.01.001.
[8] J.P. Diebold, A review of the chemical and physical mechanisms of the storage stability of fast pyrolysis bio-oils, Natl. Renew. Energy Lab. (2000). NREL/SR-570-27613.
[9] Q. Wang, X. Li, K. Wang, Y. Zhu, S. Wang, Commercialization and challenges for the next generation of biofuels: biomass fast pyrolysis, 978-1-4244-4813-5/10, Asia-Pac. Power Energy Eng. Conf. (2010), https://doi.org/10.1109/APPEEC.2010.5448437.
[10] H.S. Choi, Y.S. Choi, H.C. Park, The influence of fast pyrolysis condition on biocrude-oil yield and homogeneity, Korean J. Chem. Eng. 27 (4) (2010) 1164–1169, https://doi.org/10.1007/s11814-010-0213-8.
[11] D. Meier, A. Oasmaa, G.V.C. Peacocke, Properties of fast pyrolysis liquids: satus of test methods, Fast Pyrolysis Biomass-.-.: A Handb. 1 (1999) 75–91, https://doi.org/10.1007/978-94-009-1559-6_31.
[12] A.V. Bridgwater, G.V.C. Peacocke, Fast pyrolysis processes for biomass, Renew. Sustain. Energy Rev. 4 (1) (2000) 1–73, https://doi.org/10.1016/S1364-0321(99)00007-6.

[13] J. Wang, X. Ku, J. Lin, S. Yang, Impact of the Reactor Structure on Biomass Pyrolysis in Fluidized-Bed Reactors: A Coarse-Grained CFD-DEM Study, Energy Fuels 35 (12) (2021) 10035–10050, https://doi.org/10.1021/acs.energyfuels.1c01107.

[14] B. Urban, Y. Shirazi, B. Maddi, S. Viamajala, S. Varanasi, Pirólise instantânea de biomassa oleaginosa em um reator de leito fluidizado, Energy Fuels 31 (2017) 8326–8334, https://doi.org/10.1021/acs.energyfuels.7b01306.

[15] R.H. Venderbosch, W. Prins, Fast pyrolysis technology development, Biofuels, Bioprod. Bioref. 4 (2010) 178–208, https://doi.org/10.1002/bbb.205.

[16] K. Moser, E. Wopienka, C. Pfeifer, M. Schwarz, I. Sedlmayer, W. Haslinger, Screw reactors and rotary kilns in biochar production – a comparative review, J. Anal. Appl. Pyrolysis 174 (2023) 106112, https://doi.org/10.1016/j.jaap.2023.106112.

[17] T.R. Brown, M. Mark, M.M. Wright, R.C. Brown, Estimating profitability of two biochar production scenarios: slow pyrolysis vs fast pyrolysis, Biofuels, Bioprod. Bioref. 5 (2011) 54–68, https://doi.org/10.1002/bbb.254.

[18] P. Giudicianni, G. Cardone, G. Sorrentino, R. Ragucci, Hemicellulose, cellulose and Lignin interactions on Arundo donax steam assisted pyrolysis, J. Anal. Appl. Pyrolysis 110 (2014) 138–146, https://doi.org/10.1016/j.jaap.2014.08.014.

[19] S.V. Vassilev, D. Baxter, L.K. Andersen, C.G. Vassileva, T.J. Morgan, An overview of the organic and inorganic phase composition of biomass, Fuel 94 (2012) 1–33, https://doi.org/10.1016/j.fuel.2011.09.030.

[20] A.V. Bridgwater, Renewable fuels and chemicals by thermal processing of biomass, Chem. Eng. J. v. 91 (n. 2-3) (2003) 87–102, https://doi.org/10.1016/s1385-8947(02)00142-0.

[21] M. Sharma, J. Singh, C. Baskar, A. Kumar, A comprehensive review of renewable energy production from biomass-derived bio-oil, Biotechnologia v. 100 (n. 2) (2019) 179–194, https://doi.org/10.5114/bta.2019.85323.

[22] A.J. Ridout, M. Carrier, F.-X. Collard, J. Görgens, Energy conversion assessment of vacuum, slow and fast pyrolysis processes for low and high ash paper waste sludge, Energy Convers. Manag. v. 111 (2016) 103–114, https://doi.org/10.1016/j.enconman.2015.12.043.

[23] R. Moya, A. Rodríguez-Zúniga, A. Puente-Urbina, Thermogravimetric and devolatilisation analysis for five plantation species: effect of extractives, ash compositions, chemical compositions and energy parameters, Thermochim. Acta 647 (2017) 36–46, https://doi.org/10.1016/j.tca.2016.11.014.

[24] J.E. White, W.J. Catallo, B.L. Legendre, Biomass pyrolysis kinetics: a comparative critical review with relevant agricultural residue case studies, J. Anal. Appl. Pyrolysis v. 91 (n. 1) (2011) 1–33, https://doi.org/10.1016/j.jaap.2011.01.004.

[25] A.K. Burnham, X. Zhou, L.J. Broadbelt, Critical review of the global chemical kinetics of cellulose thermal decomposition, https://doi-org.ez10.periodicos.capes.gov.br/, Energy Fuels 29 (2015) 2906–2918, https://doi.org/10.1021/acs.energyfuels.5b00350.

[26] S. Hameed, A. Sharma, V. Pareek, H.W. Wu, Y. Yu, A review on biomass pyrolysis models: kinetic, network and mechanistic models, Biomass-.-. Bioenergy 123 (2019) 104–122, https://doi.org/10.1016/j.biombioe.2019.02.008.

[27] Y.M. Ding, B.Q. Huang, K.Y. Li, W.Z. Du, K.H. Lu, Y.S. Zhang, Thermal interaction analysis of isolated hemicellulose and cellulose by kinetic parameters during biomass pyrolysis, Energy 195 (2020) 117010, https://doi.org/10.1016/j.energy.2020.117010.

[28] K.N. Yogalakshmi, T. Poornima Devi, P. Sivashanmugam, S.B. Kavitha, R. Yukesh Kannah, S. Varjani, S. AdishKumar, G. Kumar, J. Rajesh Banu, Lignocellulosic biomass-based pyrolysis: a comprehensive review, Chemosphere 286 (2022) 131824, https://doi.org/10.1016/j.chemosphere.2021.131824.

[29] E. Anas, R. Ekman, B. Holmbom, Composition of nonpolar extractives in bark of Norway spruce and Scots pine, J. Wood Chem. Technol. 3 (1983) 119–130, https://doi.org/10.1080/02773818308085155.

[30] D. Mohan, C.U. Pittman, P.H. Steele, Pyrolysis of Wood/Biomass for Bio-oil: A critical review, https://doi-org.ez10.periodicos.capes.gov.br/, Energy Fuels 20 (3) (2006) 848–889, https://doi.org/10.1021/ef0502397.

[31] X.-j Guo, S.-r Wang, K.-g Wang, Q. Liu, Z.-y Luo, Influence of extractives on mechanism of biomass pyrolysis, J. Fuel Chem. Technol. 38 (1) (2010) 42–46, https://doi.org/10.1016/S1872-5813(10)60019-9.

[32] P.E.A. Debiagi, C. Pecchi, G. Gentile, A. Frassoldati, A. Cuoci, T. Faravelli, E. Ranzi, Extractives extend the applicability of multistep kinetic scheme of biomass pyrolysis, Energy Fuels v. 29 (n. 10) (2015) 6544–6555, https://doi.org/10.1021/acs.energyfuels.5b01753.

[33] L.M. Araujo, M.A. Santos, S.T. Brandao, S.B. Lima, C.A.M. Pires, Study of the pectin influence on bio-oil produced from sisal residue pyrolysis, J. Anal. Appl. Pyrolysis 170 (2023) 105906, https://doi.org/10.1016/j.jaap.2023.105906.

[34] T.Q.S. Sampaio, F.S. Cunha, L.M.A. Campos, C.A.M. Pires, Evaluation of the influence of hydrophilic extractives on the formation of bio-oil from the micro-pyrolysis of biomass waste, J. Anal. Appl. Pyrolysis 178 (2024) 106417, https://doi.org/10.1016/j.jaap.2024.106417.

[35] Y. Peng, X. Tang, R. Xuan, L. Wang, L. Dai, L. Zhang, F. Liao, H. Li, X. Li, Y. Shen, Y. Su, H. Wang, Analysis of pyrolysis behaviors of biomass extractives via non-linear stepwise heating program based on Gaussian multi-peak fitting of differential thermogravimetric curve, Thermochim. Acta 702 (2021) 178976, https://doi.org/10.1016/j.tca.2021.178976.

[36] D. Pradhan, R.K. Singh, H. Bendu, R. Mund, Pyrolysis of Mahua seed (Madhuca indica) – Production of biofuel and its characterization, Energy Convers. Manag. 108 (2016) 529–538, https://doi.org/10.1016/j.enconman.2015.11.042.

[37] N. Sellin, D.R. Krohl, C. Marangoni, O. Souza, Oxidative fast pyrolysis of banana leaves in fluidized bed reactor, Renew. Energy 96 (2016) 56–64, https://doi.org/10.1016/j.renene.2016.04.032.

[38] S.R. Wang, G.X. Dai, H.P. Yang, Z.Y. Luo, Lignocellulosic biomass pyrolysis mechanism: a state-of-the-art review, Prog. Energy Combust. Sci. 62 (2017) 33–86, https://doi.org/10.1016/j.pecs.2017.05.004.

[39] S.Q. Wan, N. Zheng, J. Zhang, J. Wang, Role of neutral extractives and inherent active minerals in pyrolysis of agricultural crop residues and bio-oil formations, Biomass-.-. Bioenergy 122 (2019) 53–62, https://doi.org/10.1016/j.biombioe.2019.01.010.

[40] Y.Y. Qiao, F.F. Xu, X. Ming, S.W. Feng, Y.Y. Ji, Y. Jiang, J. Li, B. Wang, Y.Y. Tian, Valorization of vegetable waste via pyrolysis: thermal behavior, volatiles release, and products analysis from its extractives, Energy Fuels 34 (2020) 1896–1907, https://doi.org/10.1021/acs.energyfuels.9b03970.

[41] M. Tajmirriahi, K. Karimi, R. Kumar, Effects of pinewood extractives on bioconversion of pinewood, Fuel 283 (2021) 119302, https://doi.org/10.1016/j.fuel.2020.119302.

[42] J. Blondeau, H. Jeanmart, Biomass pyrolysis at high temperatures: Prediction of gaseous species yields from an anisotropic particle, Biomass-.-. Bioenergy 41 (2012) 107–121, https://doi.org/10.1016/j.biombioe.2012.02.016.

[43] A.I. Ferreiro, P. Giudicianni, C.M. Grottola, M. Rabacal, M. Costa, R. Ragucci, Unresolved issues on the kinetic modeling of pyrolysis of woody and nonwoody biomass fuels, Energy Fuels 31 (2017) 4035–4044, https://doi.org/10.1021/acs.energyfuels.6b03445.

[44] Q. Lu, X.-L. Yang, X.-F. Zhu, Analysis on chemical and physical properties of bio-oil pyrolyzed from rice husk, J. Anal. Appl. Pyrolysis v. 82 (n. 2) (2008) 191–198, https://doi.org/10.1016/j.jaap.2008.03.003.

[45] A. Oasmaa, E. Kuoppala, S. Gust, Y. Solantausta, Fast pyrolysis of forestry residue. 1. effect of extractives on phase separation of pyrolysis liquids, Energy e Fuels v. 17 (n. 1) (2002) 1–12, https://doi.org/10.1021/ef020088x.

[46] M. Melzer, J. Blin, A. Bensakhria, J. Valette, F. Broust, Pyrolysis of extractive rich agroindustrial residues, J. Anal. Appl. Pyrolysis v. 104 (2013) 448–460, https://doi.org/10.1016/j.jaap.2013.05.027.

[47] A. Oasmaa, E. Kuoppala, Y. Solantausta, Fast pyrolysis of forestry residue. 2. physicochemical composition of product liquid, Energy Fuels v. 17 (2003) 433–443, https://doi.org/10.1021/ef020206g.

[48] A. Oasmaa, T. Sundqvist, E. Kuoppala, M. Garcia-Perez, Y. Solantausta, C. Lindfors, V. Paasikallio, Controlling the phase stability of biomass fast pyrolysis bio-oils, Energy Fuels v. 29 (2015) 4373–4381, https://doi.org/10.1021/acs.energyfuels.5b00607.

[49] T. Ohra-Aho, M. Ghalibaf, R. Alén, C. Lindfors, A. Oasmaa, Analysis of lipophilic extractives from fast pyrolysis bio-oils, Energy Fuels v. 36 (2022) 5797–5804, https://doi.org/10.1021/acs.energyfuels.1c04325.

[50] A. Korus, A. Szlęka, A. Samson, Physicochemical properties of biochars prepared from raw and acetoneextracted pine wood, Fuel Process. Technol. 185 (2019) 106–116, https://doi.org/10.1016/j.fuproc.2018.12.004.

[51] M. Poletto, A.J. Zattera, M.M.C. Forte, R.M.C. Santana, Thermal decomposition of wood: Influence of wood components and cellulose crystallite size, Bioresour. Technol. 109 (2012) 148–153, https://doi.org/10.1016/j.biortech.2011.11.122.

[52] J. Piskorz, D. Radlein, D. Scott, Pretreatment of wood and cellulose for production of sugars by fast pyrolysis, J. Anal. Appl. Pyrolysis 16 (2) (1989) 127–142, https://doi.org/10.1016/0165-2370(89)85012-0.

[53] V. Dhyani, T. Bhaskar, A comprehensive review on the pyrolysis of lignocellulosic biomass, Renew. Energy v. 129 (2018) 695–716, https://doi.org/10.1016/j.renene.2017.04.035.

[54] C.A. Mullen, A.A. Boateng, N.M. Goldberg, I.M. Lima, D.A. Laird, K.B. Hicks, Bio-oil and bio-char production from corn cobs and stover by fast pyrolysis, Biomass-.-. Bioenergy 34 (2010) 67–74, https://doi.org/10.1016/j.biombioe.2009.09.012.

[55] P. Ghorbannezhad, F. Kool, H. Rudi, S. Ceylan, Sustainable production of value-added products from fast pyrolysis of palm shell residue in tandem micro-reactor and pilot plant, Renew. Energy 145 (2020) 663–670, https://doi.org/10.1016/j.renene.2019.06.063.

[56] B. Gomez-Monedero, F. Bimbela, J. Arauzo, J. Faria, M.P. Ruiz, Pyrolysis of Red Eucalyptus, Camelina Straw, and Wheat Straw in an Ablative Reactor, Energy Fuels 29 (2015) 1766–1775, https://doi.org/10.1021/ef5026054.

[57] L. Cheng, S. Adhikari, Z. Wang, Y. Ding, Characterization of bamboo species at different ages and bio-oil production, J. Anal. Appl. Pyrolysis 116 (2015) 215–222, https://doi.org/10.1016/j.jaap.2015.09.008.

[58] W. Treedet, R. Suntivarakorn, Design and operation of a low cost bio-oil fast pyrolysis from sugarcane bagasse on circulating fluidized bed reactor in a pilot plant, Fuel Process. Technol. 179 (2018) 17–31, https://doi.org/10.1016/j.fuproc.2018.06.006.

[59] A.E. Putun, E. Apaydın, E. Putun, Rice straw as a bio-oil source via pyrolysis and steam pyrolysis, Energy 29 (2004) 2171–2180, https://doi.org/10.1016/j.energy.2004.03.020.

[60] T. Rout, D. Pradhan, R.K. Singh, Kumari, N. Exhaustive study of products obtained from coconut shell pyrolysis, J. Environ. Chem. Eng. 4 (2016) 3696–3705, https://doi.org/10.1016/j.jece.2016.02.024.

[61] Y. Wang, L. Wu, C. Wang, J. Yu, Z. Yang, Investigating the influence of extractives on the oil yield and alkane production obtained from three kinds of biomass via deoxy-liquefaction, Bioresour. Technol. v. 102 (n. 14) (2011) 7190–7195, https://doi.org/10.1016/j.biortech.2011.04.060.

[62] S.I. Yang, M.S. Wu, C.Y. Wu, Application of biomass fast pyrolysis part I: pyrolysis characteristics and products, Energy v. 66 (2014) 162–171, https://doi.org/10.1016/j.energy.2013.12.063.

[63] A. Oasmaa, Y. Solantausta, V. Arpiainen, E. Kuoppala, K. Sipila, Fast Pyrolysis Bio-Oils from Wood and Agricultural Residues, Energy Fuels 24 (2010) 1380–1388, https://doi.org/10.1021/ef901107f.

[64] Silvério, F.O. Caracterização de extrativos de madeira de eucalyptus e depósitos de pitch envolvidos na fabricação. 180 f. Tese (Doutorado) - Departamento de Química, Universidade Federal de Minas Gerais, Belo Horizonte, 2008.
[65] A.J.V. Zanuncio, J.L. Colodette, F.J.B. Gomes, A.De.C.O. Carneiro, B.R. Vital, Composição química da madeira de eucalipto com diferentes níveis de desbaste, Ciência Florest. v. 23 (n. 4) (2013) 755–760, https://doi.org/10.5902/1980509812359.
[66] H.F. dos Santos, Análise conformacional de modelos de lignina, Química Nova v. 24 (n. 4) (2001) 480–490, https://doi.org/10.1590/S0100-40422001000400009.
[67] Rueda, S.M.G. Pré-tratamento e Hidrólise Enzimática do Bagaço de Cana-de-açúcar. 2010. 130 f. Dissertação (Mestrado em Engenharia Química) – Universidade Estadual de Campinas, Campinas, SP, 2010.
[68] P. Tanger, J.L. Field, C.E. Jahn, M.W. Defoort, J.E. Leach, Biomass for thermochemical conversion: targets and challenges, Front. Plant Sci. v. 4 (2013) 1–20, https://doi.org/10.3389/fpls.2013.00218.
[69] F. Masarin, D.B. Gurpilhares, D.C. Baffa, Chemical composition and enzymatic digestibility of sugarcane clones selected for varied lignin content, Biotechnol. Biofuels 4 (2011) 55, https://doi.org/10.1186/1754-6834-4-55.
[70] Philippini, R.R. Variedades híbridas de bagaço de cana-de-açúcar: caracterização química e hidrólise enzimática em condições de pré-tratamento difrenciadas. 2012. 82 f. Dissertação (Mestrado em Biotecnologia Industrial) - Escola de Engenharia de Lorena da Universidade de São Paulo, 2012.
[71] N.V. Hoang, A. Furtado, L. Donnan, E.C. Keeffe, F.C. Botha, R.J. Henry, High-Throughput profiling of the fiber and sugar composition of sugarcane biomass, Bioenergy Res. v. 10 (n. 2) (2017) 400–416, https://doi.org/10.1007/s12155-016-9801-8.
[72] C. Kuchelmeister, S. Bauer, Rapid small-scale determination of extractives in biomass, Bioenergy Res. v. 8 (n. 1) (2015) 68–76, https://doi.org/10.1007/s12155-014-9493-x.
[73] L. Canilha, V.T.O. Santos, G.J.M. Rocha, J.B.A. Silva, M. Giulietti, S.S. Silva, M.G. A. Felipe, A. Ferraz, A.M.F. Milagres, W. Carvalho, A study on the pretreatment of a sugarcane bagasse sample with dilute sulfuric acid, J. Ind. Microbiol. Biotechnol. v. 38 (n. 9) (2011) 1467–1475, https://doi.org/10.1007/s10295-010-0931-2.
[74] U.S. Orlando, A.U. Baes, W. Nishijima, M. Okada, Preparation of agricultural residue anion exchangers and its nitrate maximum adsorption capacity, Chemosphere v. 48 (n. 10) (2002) 1041–1046, https://doi.org/10.1016/s0045-6535(02)00147-9, 014-9493-x.
[75] J. Gominho, A. Lourenço, A.V. Marques, H. Pereira, An extensive study on the chemical diversity of lipophilic extractives from Eucalyptus globulus wood, Phytochemistry v. 180 (2020) 112520, https://doi.org/10.1016/j.phytochem.2020.112520.
[76] M.B. Folgueras, J.M. Gomez-Martín, M.A. Diez, How height-related variations in hybrid poplars affect composition and pyrolytic behaviour: The key role of lignin maturity during woody-biomass pyrolysis, J. Anal. Appl. Pyrolysis 169 (2023) 105861, https://doi.org/10.1016/j.jaap.2023.105861.
[77] S.D. Rabiu, M. Auta, A.S. Kovo, An upgraded bio-oil produced from sugarcane bagasse via the use of HZSM-5 zeolite catalyst, Egypt. J. Pet. v. 27 (n. 4) (2018) 589–594, https://doi.org/10.1016/j.ejpe.2017.09.001.
[78] T. Kan, V. Strezov, T. Evans, J. He, R. Kumar, Q. Lu, Catalytic pyrolysis of lignocellulosic biomass: a review of variations in process factors and system structure, Renew. Sustain. Energy Rev. v. 134 (2020) 110305, https://doi.org/10.1016/j.rser.2020.110305.
[79] R.P. Rodgers, T.M. Schaub, A.G. Marshall, Petroleomics: MS returns to its roots, Anal. Chem., Wash. v. 77 (n. 1) (Jan. 2005) 20A–27A, https://doi.org/10.1021/ac053302y.
[80] Z. Kaczor, Z. Buliński, S. Werle, Modelling approaches to waste biomass pyrolysis: a review, Renew. Energy v. 159 (2020) 427–443, https://doi.org/10.1016/j.renene.2020.05.110.
[81] T. Kan, V. Strezov, T.J. Evans, Lignocellulosic biomass pyrolysis: a review of product properties and effects of pyrolysis parameters, Renew. Sustain. Energy Rev. v. 57 (2016) 1126–1140, https://doi.org/10.1016/j.rser.2015.12.185.
[82] Z. Guo, K. Zhang, Q. Zhang, L. Fu, Z. Liu, Z. Kong, L. Wang, C. Liu, L. Hua, B. Li, Tobacco fractionation and its effects on pyrolysis chemistry, J. Anal. Appl. Pyrolysis 167 (2022) 105650, https://doi.org/10.1016/j.jaap.2022.105650.
[83] C. Fushimi, S. Katayama, A. Tsutsumi, Elucidation of interaction among cellulose, lignin and xylan during tar and gas evolution in steam gasification, J. Anal. Appl. Pyrolysis 86 (2009) 82–89, https://doi.org/10.1016/j.jaap.2009.04.008.
[84] T. Hosoya, H. Kawamoto, S. Saka, Cellulose-hemicellulose and cellulose-lignin interactions in wood pyrolysis at gasification temperature, J. Anal. Appl. Pyrolysis 80 (2007) 118–125, https://doi.org/10.1016/j.jaap.2007.01.006.
[85] H.L. Ornaghi Jr, M. Poletto, A.J. Zattera, S.C. Amico, Correlation of the thermal stability and the decomposition kinetics of six different vegetal fibers, Cellulose 21 (2014) 177–188, https://doi.org/10.1007/s10570-013-0094-1.
[86] B.V. L'vov, The physical approach to the interpretation of the kinetics and mechanisms of thermal decomposition of solids: the state of the art, Thermochim. Acta 373 (2001) 97–124, https://doi.org/10.1016/S0040-6031(01)00507-X.
[87] A. Demirbas, Estimating of structural composition of wood and non-wood biomass samples, Energy Sources 27 (8) (2005) 761–767, https://doi.org/10.1080/00908310490450971.
[88] E. Mészáros, E. Jakab, G. Várhegyi, TG/MS, Py-GC/MS and THM-GC/MS study of the composition and thermal behavior of extractive components of Robinia pseudoacacia, J. Anal. Appl. Pyrolysis 79 (1/2) (2007) 61–70, https://doi.org/10.1016/j.jaap.2006.12.007.
[89] L. Dai, Y. Wang, Y. Liu, C. He, R. Ruan, Z. Yu, L. Jiang, Z. Zeng, Q. Wu, A review on selective production of value-added chemicals via catalytic pyrolysis of lignocellulosic biomass, Sci. Total Environ. v. 749 (2020) 142386, https://doi.org/10.1016/j.scitotenv.2020.142386.
[90] M. Zheng, Z. Wang, X. Li, X. Qiao, W. Song, L. Guo, Initial reaction mechanisms of cellulose pyrolysis revealed by ReaxFF molecular dynamics, Fuel v. 177 (2016) 130–141, https://doi.org/10.1016/j.fuel.2016.03.008.
[91] S.D. Stefanidis, K.G. Kalogiannis, E.F. Iliopoulou, C.M. Michailof, P.A. Pilavachi, A.A. Lappas, A study of lignocellulosic biomass pyrolysis via the pyrolysis of cellulose, hemicellulose and lignin, J. Anal. Appl. Pyrolysis v. 105 (2014) 143–150, https://doi.org/10.1016/j.jaap.2013.10.013.
[92] J.K. Lindstrom, J. Proano-Aviles, P.A. Johnston, C.A. Peterson, J.S. Stansell, R. C. Brown, Competing reactions limit levoglucosan yield during fast pyrolysis of cellulose, Green. Chem. v. 21 (n. 1) (2019) 178–186, https://doi.org/10.1039/c8gc03461c.
[93] H. Yang, S. Li, B. Liu, Y. Chen, J. Xiao, Z. Dong, M. Gong, H. Chen, Hemicellulose pyrolysis mechanism based on functional group evolutions by twodimensional perturbation correlation infrared spectroscopy, Fuel v. 267 (2020) 117302, https://doi.org/10.1016/j.fuel.2020.117302.
[94] X. Zhou, W. Li, R. Mabon, L.J. Broadbelt, A Critical Review on Hemicellulose Pyrolysis, Energy Technol. v. 5 (n. 1) (2017) 52–79, https://doi.org/10.1002/ente.201600327.
[95] S. Wang, B. Ru, H. Lin, W. Sun, Pyrolysis behaviors of four O-acetyl-preserved hemicelluloses isolated from hardwoods and softwoods, Fuel v. 150 (2015) 243–251, https://doi.org/10.1016/j.fuel.2015.02.045.
[96] X. Yang, Y. Zhao, W. Li, R. Li, Y. Wu, Unveiling the Pyrolysis Mechanisms of Hemicellulose: experimental and theoretical studies, Energy Fuels v. 33 (n. 5) (2019) 4352–4360, https://doi.org/10.1021/acs.energyfuels.9b00482.
[97] Q. Liu, S.R. Wang, Y. Zheng, Z.Y. Luo, K.F. Cen, Mechanism study of wood lignin pyrolysis by using TG-FTIR analysis, J. Anal. Appl. Pyrolysis 82 (1) (2008) 170–177, https://doi.org/10.1016/j.jaap.2008.03.007.
[98] S.L. Zuo, J. Yu, S.W. Chen, Effect of pyrolysis temperature on pyrolysate during fast pyrolysis of industrial lignin made by acid precipitation, J. Fuel Chem. Technol. 36 (2) (2008) 144–148, https://doi.org/10.1016/S1872-5813(10)60019-9.
[99] M. Balat, Mechanisms of thermochemical biomass conversion processes. part 1: reactions of pyrolysis, Energy Sources, Part A: Recovery, Util., Environ. Eff. v. 30 (n. 7) (2008) 620–635, https://doi.org/10.1080/15567030600817258.
[100] S. Chu, A.V. Subrahmanyam, G.W. Huber, The pyrolysis chemistry of a β-O-4 type oligomeric lignin model compound, Green. Chem. v. 15 (n. 1) (2013) 125–136, https://doi.org/10.1039/c2gc36332a.
[101] W. Mu, H. Ben, A. Ragauskas, Y. Deng, Lignin Pyrolysis components and upgrading-technology review, Bioenergy Res. v. 6 (n. 4) (2013) 1183–1204, https://doi.org/10.1007/s12155-013-9314-7.
[102] G. Jiang, D.J. Nowakowski, A.V. Bridgwater, Effect of the temperature on the composition of lignin pyrolysis products, Energy Fuels v. 24 (n. 8) (2010) 4470–4475, https://doi.org/10.1021/ef100363c.
[103] P.R. Patwardhan, R.C. Brown, B.H. Shanks, Understanding the Fast Pyrolysis of Lignin, Chemsuschem v. 4 (n. 11) (2011) 1629–1636, https://doi.org/10.1002/cssc.201100133.
[104] A. Anca-Couce, Reaction mechanisms and multi-scale modelling of lignocellulosic biomass pyrolysis, Prog. Energy Combust. Sci. v. 53 (2016) 41–79, https://doi.org/10.1016/j.pecs.2015.10.002.
[105] U. Sen, B. Esteves, H. Pereira, Pyrolysis and extraction of bark in a biorefineries context: a critical review, Energies v. 16 (2023) 4848, https://doi.org/10.3390/en16134848.
[106] A. Demirbas, Mechanisms of liquefaction and pyrolysis reactions of biomass, Energy Convers. Manag. 41 (2000) 633–646, https://doi.org/10.1016/S0196-8904(99)00130-2.
[107] Y. Tian, J. Zhang, W. Zuo, L. Chen, Y. Cui, T. Tan, Nitrogen conversion in relation to NH3 and HCN during microwave pyrolysis of sewage sludge, Environ. Sci. Technol. 47 (2013) 3498–3505, https://doi.org/10.1021/es304248j.
[108] Z. Chen, E. Leng, Y. Zhang, A. Zheng, Y. Peng, X. Gong, Y. Huang, Y. Qiao, Pyrolysis characteristics of tobacco stem after different solvent leaching treatments, J. Anal. Appl. Pyrolysis 130 (2017) 350–357, https://doi.org/10.1016/j.jaap.2017.12.009.
[109] M. Pagano, H. Hernando, J. Cueto, P.L. Cruz, J. Dufour, In'es, I. Moreno, D. P. Serrano, Insights on the acetic acid pretreatment of wheat straw: Changes induced in the biomass properties and benefits for the bio-oil production by pyrolysis, Chem. Eng. J. 454 (2023) 140206, https://doi.org/10.1016/j.cej.2022.140206.
[110] Y. Huang, D.T. Sekyere, J. Zhang, Y. Tian, Fast pyrolysis behaviors of biomass with high contents of ash and nitrogen using TG-FTIR and Py-GC/MS, J. Anal. Appl. Pyrolysis 170 (2023) 105922, https://doi.org/10.1016/j.jaap.2023.105922.
[111] H. Yang, R. Yan, H. Chen, D. Lee, C. Zheng, Characteristics of hemicellulose, cellulose and Lignin pyrolysis, Fuel 86 (2007) 1781–1788, https://doi.org/10.1016/j.fuel.2006.12.013.
[112] X. Shi, J. Wang, A comparative investigation into the formation behaviors of char, liquids and gases during pyrolysis of pinewood and lignocellulosic components, Bioresour. Technol. 170 (2014) 262–269, https://doi.org/10.1016/j.biortech.2014.07.110.
[113] A. Demirbas, Mechanisms of liquefaction and pyrolysis reactions of biomass, Energy Convers. Manag. 41 (2000) 633–646, https://doi.org/10.1016/S0196-8904(99)00130-2.
[114] Y. Dai, J. Xu, L. Zhu, J. Jiang, Y. Zhou, G. Zhou, Mechanism study on the effect of glycerol addition on tobacco pyrolysis, J. Anal. Appl. Pyrolysis 157 (2021) 105183, https://doi.org/10.1016/j.jaap.2021.105183.

[115] T. Kan, V. Strezov, T. Evans, J. He, R. Kumar, Q. Lu, Catalytic pyrolysis of lignocellulosic biomass: a review of variations in process factors and system structure, Renew. Sustain. Energy Rev. v. 134 (2020) 110305, https://doi.org/10.1016/j.rser.2020.110305.
[116] M. He, D. Mourant, R. Gunawan, C. Lievens, X.S. Wang, K. Ling, J. Bartle, C.Z. Li, Yield and properties of bio-oil from the pyrolysis of mallee leaves in a fluidised-bed reactor, Fuel 102 (2012) 506–513, https://doi.org/10.1016/j.fuel.2012.07.003.
[117] U. Raisanen, I. Pitkanen, H. Halttunen, M. Hurtta, Formation of the main degradation compounds from arabinose, xylose, mannose and arabinitol during pyrolysis, J. Therm. Anal. Calor. 72 (2003) 481–488, https://doi.org/10.1023/A:1024557011975.
[118] S.L. Wu, D.K. Shen, J. Hu, H.Y. Zhang, R. Xiao, Role of beta-O-4 glycosidic bond on thermal degradation of cellulose, J. Anal. Appl. Pyrolysis 119 (2016) 147–156, https://doi.org/10.1016/j.jaap.2016.03.006.
[119] D. Che, L. Wang, H. Liu, B. Sun, S. Guo, Effects of lipids on the conversion of nitrogen during protein pyrolysis: a model study, J. Environ. Chem. Eng. 10 (2022) 108570, https://doi.org/10.1016/j.jece.2022.108570.